Rubel Roy

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Professional Summary

Data Science postgraduate with hands-on experience in Python, SQL, statistical analysis, and machine learning. Currently interning with the Government of West Bengal, collaborating in a three-member team on an AI-driven project. Strong analytical and communication skills, with a focus on turning data into actionable business insights.

Education

St.Xavier's College (Autonomous),Kolkata — Kolkata, India
M.Sc. in Data Science; CGPA: 7.17 — *2025 – 2027*

Gurudas College — Kolkata, India
B.Sc. in Physics; CGPA: 7.55 — *2020 – 2024*

Technical Skills

Generative AI & LLMs: Local LLMs (Ollama: Llama 3.2, Qwen), Fine-tuning (LoRA, QLoRA, PEFT), Hugging Face (Transformers, Datasets), RAG (Chunking, Embeddings, Vector DB, Reranking, RAG Evaluation), Model Benchmarking
Machine Learning & NLP: Classification, Regression, Clustering, Random Forest, XGBoost, Feature Engineering, Model Evaluation, SHAP, Transformers, LSTM/GRU, CNN
Statistics & Analytics: EDA, Hypothesis Testing, Regression Analysis, Predictive Analytics, Customer Segmentation, RFM, Cohort Analysis
Languages & Tools: Python, SQL, R, Jupyter, FastAPI, Flask, Streamlit, Git, GitHub

Experience

Centre of Excellence on Data Science and Machine Learning, WEBEL — Aug 2026 – Present
AI/ML Intern — *Kolkata, India*

- Building an AI-powered video generation pipeline using FastAPI to create Bengali awareness videos from government scheme documents.
- Integrating Gemini APIs for information extraction, script generation, and scene-wise prompt creation.
- Implementing LLM-based prompt validation using GPT-OSS-120B and Qwen3.8-27B for factual consistency and regional relevance.
- Automating video generation and assembly using Google Omni Flash Model, MoviePy, and FFmpeg.

Projects

Local RAG Resume Assistant | *Python, LangChain, ChromaDB, Ollama, Streamlit* | GitHub

- Built an offline RAG pipeline using Ollama (Llama 3.2 3B, nomic-embed-text) and ChromaDB for semantic indexing and retrieval of PDF/DOCX resumes.
- Developed an ATS scoring engine evaluating resumes across four dimensions: XYZ impact, keyword relevance, structure, and tone, with a 100-point scoring system and actionable bullet-point improvements.
- Implemented streaming conversational Q&A using LangChain LCEL, generating structured career recommendations and resume-based marketability assessments.
- Built an interactive Streamlit interface with real-time response streaming, Ollama health monitoring, and automated vector store lifecycle management.

Explainable Credit Risk Assessment System | *Python, scikit-learn, XGBoost, SHAP* | GitHub

- Built an end-to-end loan-default prediction pipeline on a 32.5K-applicant dataset, handling class imbalance (78% repaid / 22% default) with XGBoost class weighting.
- Tuned XGBoost with RandomizedSearchCV (150 configurations, 5-fold CV), improving test F1 from 0.65 to 0.81 with a 0.94 ROC-AUC, outperforming Logistic Regression by 16 F1 points.
- Optimized the classification threshold to $\sim$0.695 using Precision-Recall analysis and calibrated probabilities using Platt scaling.
- Used SHAP to explain overall and individual predictions, and deployed the model via a FastAPI service with a dashboard on Render.